🚀 Backend Developer Roadmap (With Frontend Sprint)

Target: Tech Job Right Out of College (~1.3 Years Timeline)
Profile: Mech Engg, Tier 3 College, Little bit of Coding Exp, GitHub Familiar

📍 Phase 1: Frontend Sprint (Week 1–2)

> 🔥 Goal: Learn just enough frontend to build interfaces for your backend projects.

✅ Week 1: HTML + CSS + JS

- [x] HTML Tags, Forms, Inputs, Buttons (16/04/2025)
- [x] CSS Basics: Flexbox, Grid, Box Model (18/04/2025)
- [x] Responsive Design (media queries) (20/04/2025)
- [x] JavaScript Essentials:
  - [x] `let` , `const` , functions, objects, arrays (19/04/2025)
  - [x] Arrow functions, map/filter/reduce (19/04/2025)
  - [x] Async JS (Promises, Fetch) (20/04/2025)

✅ Week 2: React.js Basics

- [ ] React setup with Vite or Create React App
- [ ] Components, Props, State
- [ ] `useState` , `useEffect`
- [ ] React Router (`BrowserRouter` , `Route` , `Link`)
- [ ] Forms & input handling
- [ ] Fetch API or Axios for API calls
- [ ] Simple UI project: Blog reader or Login page
- [ ] Deploy with **Vercel**

📍 Phase 2: Backend Core (Month 1–2)

> 🎯 Goal: Understand how backend works using Node.js + Express.

✅ Week 3–4: Node.js + Express.js

- [ ] What is Node.js? How backend works?
- [ ] Setup Node.js project: `npm init`, `package.json`
- [ ] Express app setup: routes, `app.get`, `app.post`, etc.
- [ ] Middleware (`express.json()`, custom middleware)
- [ ] Error handling with middleware
- [ ] Serving static files

✅ Week 5–6: REST API + CRUD

- [ ] HTTP Verbs (GET, POST, PUT, DELETE)
- [ ] Build full CRUD API (e.g., Todos)
- [ ] Route structuring (router files)
- [ ] Status codes, JSON responses
- [ ] Postman/Thunder Client testing
- [ ] Environment Variables (`dotenv`)
- [ ] Nodemon for dev workflow

✅ Mini Projects:

- [ ] Notes API
- [ ] Bookstore API
- [ ] Deploy one on **Render** or **Railway**

📍 Phase 3: Full Backend Stack (Month 3–5)

> 🧠 Goal: Learn database, auth, real-world features, and full project workflows.

✅ Week 7–9: MongoDB + Mongoose

- [ ] Setup MongoDB Atlas
- [ ] What is NoSQL? Schema design principles
- [ ] Mongoose Models & Schemas
- [ ] Validation, Required fields, Data types
- [ ] CRUD operations with MongoDB
- [ ] Population (`ref`, `populate`)
- [ ] Connect frontend with backend

✅ Week 10–12: Auth System

- [ ] Signup/Login with JWT

- [ ] Hashing passwords with bcrypt
- [ ] Role-based access (admin/user)
- [ ] Middleware for protected routes
- [ ] Storing JWT in cookies vs headers

✅ Week 13–14: Advanced Features

- [ ] File Upload (Multer or Cloudinary)
- [ ] Email Verification with Nodemailer
- [ ] Filtering, Sorting, Pagination
- [ ] CORS setup

✅ Major Project (Full-stack):

- [ ] Blogging Platform or Job Board
  - [ ] Auth (JWT)
  - [ ] MongoDB backend
  - [ ] React frontend
  - [ ] Live deployment
- [ ] Write README with tech stack, features, how to run

📍 Phase 4: System Thinking + Portfolio (Month 6–8)

> 🎯 Goal: Polish your profile and learn intermediate backend principles.

✅ Week 15–17: Testing & System Design Basics

- [ ] API Testing with Postman Collections
- [ ] Unit Testing with Jest (intro level)
- [ ] System Design:
  - [ ] REST vs GraphQL
  - [ ] Scaling databases (vertical/horizontal)
  - [ ] Caching with Redis (basic)
  - [ ] Load Balancers, Rate Limiting

✅ Week 18–19: Portfolio Site

- [ ] Build personal portfolio using React + Tailwind
- [ ] Include:
  - [ ] About section

- [ ] Resume (PDF)
- [ ] Projects (with links)
- [ ] Contact info / LinkedIn

✅ Cleanup Checklist:

- [ ] GitHub profile: clean, pinned repos
- [ ] LinkedIn: headline, summary, links to projects
- [ ] Resume: tech stack, projects, GitHub, email

📍 Phase 5: Job Prep (Month 8–13)

> 🔥 Goal: Start applying, build network, crack interviews.

✅ Weekly Routine (Ongoing):

- [ ] 1–2 hours DSA daily (LeetCode):
  - [ ] Arrays, Strings, HashMap
  - [ ] Stack, Queue, Linked List
  - [ ] Trees, Graphs (BFS/DFS)
  - [ ] Recursion, Backtracking
- [ ] Weekend: mock interviews (Pramp, peers)

✅ Application Strategy:

- [ ] Apply on:
  - [ ] LinkedIn
  - [ ] CutShort
  - [ ] AngelList
  - [ ] Hirect
  - [ ] Internshala
- [ ] Cold email startup founders with:
  - [ ] GitHub + Portfolio + Short message

✅ Optional Skills (for bonus edge):

- [ ] PostgreSQL + Prisma ORM
- [ ] Docker basics
- [ ] GraphQL fundamentals
- [ ] Open Source contributions (EddieHub, FirstTimersOnly)

📋 Final Checklist (Before Applying)

- [ ] ✅ 3+ Projects: At least 1 major full-stack
- [ ] ✅ Deployed Projects + GitHub clean
- [ ] ✅ Resume ready + LinkedIn updated
- [ ] ✅ Personal site live with links
- [ ] ✅ Project explanation skills solid
- [ ] ✅ Consistent GitHub commits
- [ ] ✅ 300+ LeetCode problems (Easy/Medium focused)